

Wireshark Cheat Sheet for Beginners

Capture filters, display filters, and GUI tips to get started

Capture Filters

Set before you start capturing, in Capture Options. These use Berkeley Packet Filter (BPF) syntax and limit what gets captured at all.

Filter	Purpose
host 192.168.1.1	Traffic to/from a specific IP
net 192.168.1.0/24	Traffic within a subnet
port 80	Traffic on port 80 (either direction)
src port 443	Traffic from source port 443
dst host 8.8.8.8	Traffic to a specific destination
tcp / udp / icmp	Filter by protocol
tcp port 80 or tcp port 443	Multiple ports
not broadcast and not multicast	Exclude noisy broadcast/multicast
ether host aa:bb:cc:dd:ee:ff	Filter by MAC address
portrange 1000-2000	Range of ports

Combine with **and**, **or**, **not** (or &&, ||, !).

Display Filters

Used in the filter bar after capture. These use Wireshark's own syntax and only hide/show packets already captured, without discarding data.

By protocol / field:

```
http
dns
tls
tcp.port == 443
ip.addr == 192.168.1.1
ip.src == 10.0.0.5
tcp.flags.syn == 1 && tcp.flags.ack == 0 # SYN only (connection attempts)
tcp.analysis.retransmission # retransmitted packets
http.request.method == "POST"
http.response.code == 404
dns.qry.name == "example.com"
tls.handshake.type == 1 # Client Hello
```

Common operators:

Operator	Meaning
== , !=	equals / not equals

Operator	Meaning
&& , , !	and / or / not
contains	substring match, e.g. http.host contains "google"
matches	regex match, e.g. http.request.uri matches "\.php\$"
> , < , >= , <=	numeric comparison, e.g. frame.len > 1000

Useful combos:

```
ip.addr == 192.168.1.1 && tcp.port == 80
!(arp or icmp or dns) # hide chatty protocols
http.request or http.response
tcp.flags.reset == 1 # RST packets
frame contains "password" # payload search (any layer)
```

GUI Workflow Tips

- **Statistics** → **Conversations**: see all talkers, sorted by bytes/packets.
- **Statistics** → **Protocol Hierarchy**: breakdown of traffic by protocol.
- **Follow** → **TCP Stream** (right-click a packet): reconstruct a full conversation.
- **File** → **Export Objects** → **HTTP**: pull out files transferred over HTTP.
- **Right-click a field** → **Apply as Filter**: quickly build filters from packet details.
- **Edit** → **Preferences** → **Protocols** → **TLS**: point to a keylog file to decrypt TLS if you have the session keys.
- **Color coding**: packets are color-coded by type (e.g. black = errors, green = HTTP) — hover a packet to see why under View → Coloring Rules.

Keyboard Shortcuts

Shortcut	Action
Ctrl+E	Start/stop capture
Ctrl+F	Find packet
Ctrl+G	Go to packet number
Ctrl+R	Reload capture file
Ctrl+M	Mark/unmark packet
Ctrl+Alt+Shift+T	Apply as filter (selected field)

Getting Started Tips

- Pick the right interface first (Wi-Fi vs Ethernet) — if unsure, capture on "any" temporarily to see which one has traffic.
- Start with no filter, capture a few seconds of normal activity, then apply display filters to narrow down what you're looking at. Don't over-filter at capture time as a beginner — you can always filter later.
- Use **http** or **dns** as your first display filters — they're easy to read and good for learning packet structure.
- Click on a packet, then expand the panes below (Frame, Ethernet, IP, TCP, etc.) to see how each layer is structured.